Clinical Risk Assessment Report: Patient 37

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Borderline / Low Confidence
- Absolute Predicted Risk: 21.2% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.96x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Patient Age (Age), which reduced the patient's absolute risk by 2.9%. Biologically, this feature patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Clinical Sensitivity Analysis (Borderline Patient):

PREDICTION INSTABILITY WARNING: This patient's risk profile sits precisely on the borderline. Consequently, the model's prediction is highly sensitive. Minor biological variations or measurement errors in the following features would flip the final clinical prediction:

- Patient Age: A decrease would alter the prediction outcome.
- Large Zone Emphasis (CT): An increase would alter the prediction outcome.
- Large Zone Low Gray-Level Emphasis (CT): An increase would alter the prediction outcome.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: 2.68 [Top 3%] (+3.8%)**

Points to continuous, elongated bands of high-density or highly active tumor tissue.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.29 [Avg (28th %ile)] (+1.8%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 2.65 [Top <1%] (+1.7%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Minimum Tumor Density (CONVENTIONAL_HUmin) **Value: 0.48 [Avg (69th %ile)] (+1.4%)**

Reflects the least dense areas within the tumor on CT, often corresponding to cavitation, necrosis, or cystic degeneration.

3. Protective / Mitigating Factors (Reducing Risk)

Patient Age (Age) **Value: 65.0 [Avg (54th %ile)] (-2.9%)**

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: -0.4 [Bottom <1%] (-1.7%)**

Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

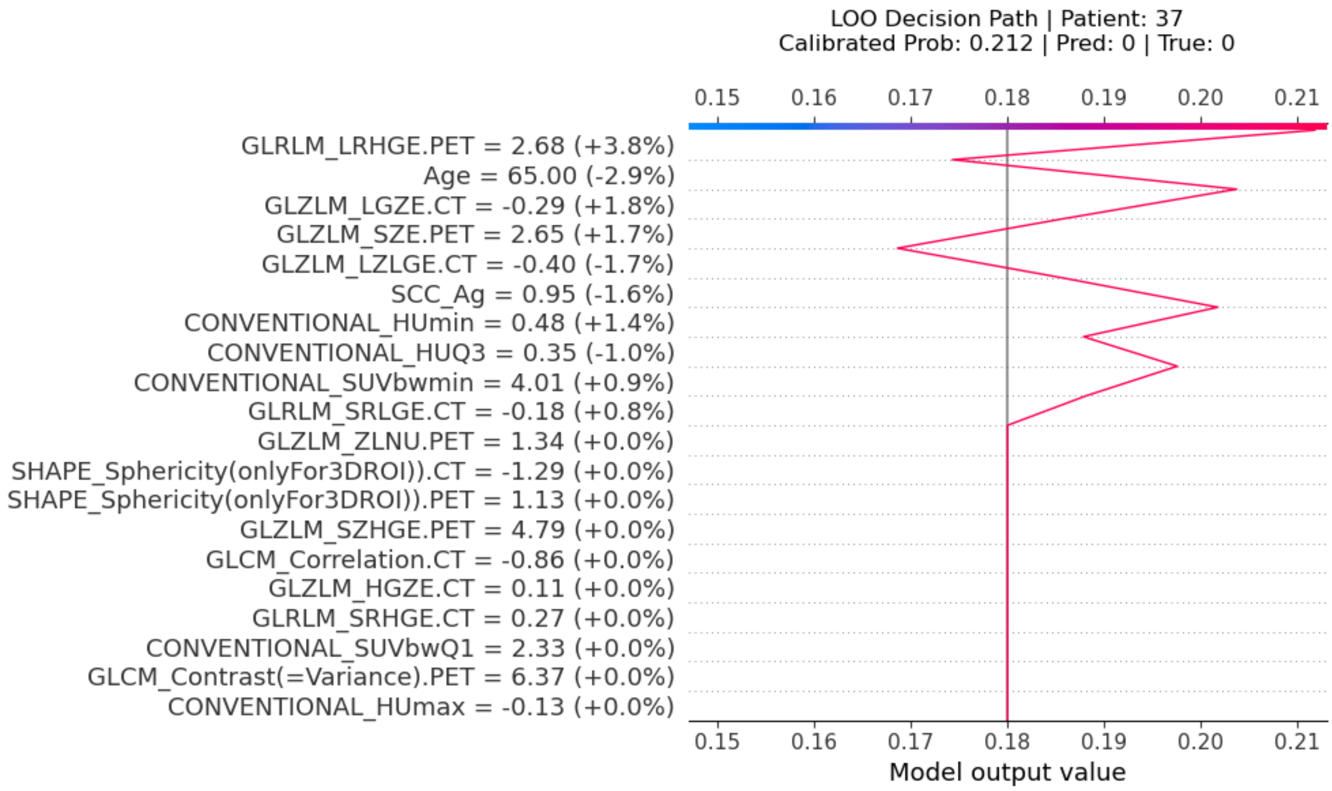
Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 0.95 [Top 18%] (-1.6%)**

Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.35 [Avg (53th %ile)] (-1.0%)**

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Machine Learning Feature Attribution (SHAP Decision Path)



Sensitivity Analysis: Feature Tipping Points

The plots below demonstrate how variations in specific clinical or radiomic features affect the model's predicted risk. The red marker indicates the patient's current clinical state. Crossing the red dashed line changes the diagnosis.